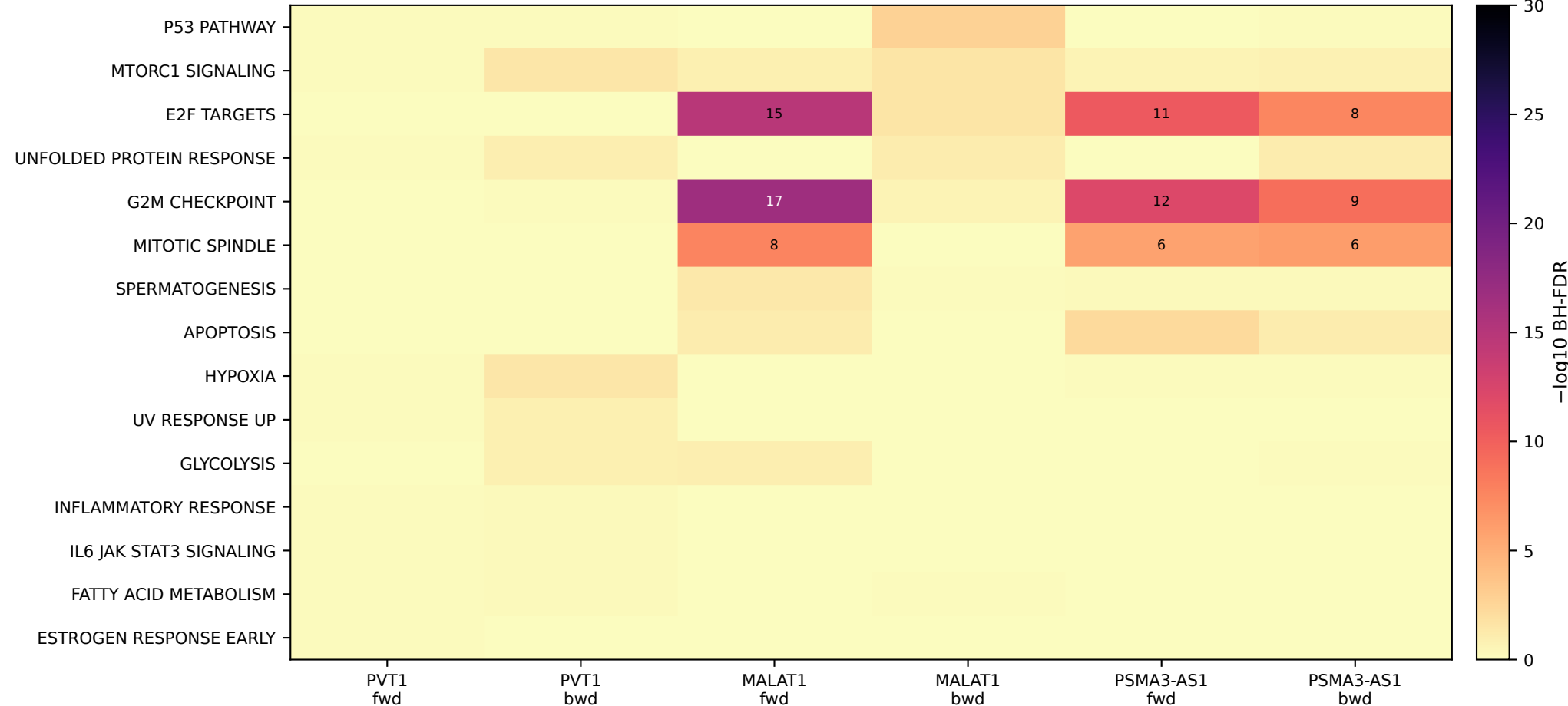


Figure 5 — Validation | Hallmark enrichment (gene programs) + permutation control (operator magnitude)

a | Hallmark over-representation in scJDO top-50 genes ($-\log_{10}$ FDR)



b | Permutation control (PVT1, n=20) empirical p = 0.143

